Week 5 formula sheet · self-study version

Forecasting — Heizer Ch. 4 · naive · moving average · exponential smoothing · seasonal index · trend · MAD / MAPE · tracking signal · for Test 3 preparation

1 · Measuring the miss error, MAD, MSE, MAPE

Error = actual − forecast (keep the sign) · **|error|** = the miss without its sign

MAD = sum of |error| ÷ number of periods

MSE = sum of error² ÷ number of periods

MAPE = sum of (|error| ÷ actual × 100 %) ÷ number of periods

2 · Naive and moving averages the level, no pattern

Naive: forecast for tonight = last night's actual

Moving average (n nights): forecast = (the last n actuals added up) ÷ n

Weighted moving average: forecast = Σ (weight × actual) ÷ Σ weights (recent nights weigh more)

forecasts in **whole rolls**; x.5 rounds up

3 · Exponential smoothing α = how nervous

New forecast = old forecast + **α** × (last actual − old forecast)

α near 0: calm, slow to react · α near 1: nervous, follows last night · start value given (e.g. last week's average)

Trend-adjusted (Holt): smooth the level with α *and* the slope with β; forecast = level + slope

4 · Seasonal index the pattern you could have used

Weekday index = average of that weekday ÷ average of all nights (two decimals; six indices add up to 6.00)

Seasonal forecast = level × index, level = last week's average (whole rolls)

a known event (festival, holiday) is calendar knowledge: adjust by hand, do not wait for the data

5 · Trend projection a straight line through the past

$\hat{y} = a + b\,x$ (x = period number, ŷ = forecast)

$b = \Sigma (x - \bar{x})(y - \bar{y}) \div \Sigma (x - \bar{x})^2$ · $a = \bar{y} - b\,\bar{x}$

Associative model: the same line on another variable (price, temperature); **r** = how close the points sit to the line, **r²** = share explained

6 · Is the forecast biased? tracking signal

Bias = the forecast sits on one side of the actuals · **RSFE** = all errors so far added up, with signs

Tracking signal = RSFE ÷ MAD (MAD = all |errors| so far ÷ periods so far)

inside **±4**: noise, leave it · outside: a signal — re-set the method (as a control chart in Week 4)

Three checks: the forecast is written before the actual is known · whole rolls, x.5 up · MAD and MAPE compare methods on the same nights; a known calendar event is adjusted by hand

Heizer, J., Render, B., & Munson, C. (2023). *Operations management: Sustainability and supply chain management* (14th ed., Global ed.). Pearson. Chapter 4.